

ECON 300 – Labour Economics

Solutions: Practice Problem Set 5

Question 1. Immigration composition

Total immigrants:

$$320,000$$

(a) Economic immigrants:

$$\frac{160,000}{320,000} \times 100 = 50\%$$

$$\boxed{50\%}$$

(b) Family-class immigrants:

$$\frac{96,000}{320,000} \times 100 = 30\%$$

$$\boxed{30\%}$$

(c) Refugees / humanitarian immigrants:

$$\frac{64,000}{320,000} \times 100 = 20\%$$

$$\boxed{20\%}$$

(d) Economic immigrants rise by 20%:

$$160,000(1.20) = 192,000$$

New total:

$$192,000 + 96,000 + 64,000 = 352,000$$

$$\boxed{352,000}$$

(e) New share of economic immigrants:

$$\frac{192,000}{352,000} \times 100 \approx 54.55\%$$

$$\boxed{54.55\%}$$

Question 2. Immigration and equilibrium wage/employment

Demand:

$$W = 32 - 0.02N$$

Supply:

$$W = 8 + 0.01N$$

(a) Initial equilibrium: set demand = supply

$$32 - 0.02N = 8 + 0.01N$$

$$24 = 0.03N$$

$$N = 800$$

Substitute into either curve:

$$W = 8 + 0.01(800) = 16$$

$$\boxed{N = 800, \quad W = 16}$$

(b) New supply:

$$W = 8 + 0.01(N - 120)$$

Simplify:

$$W = 8 + 0.01N - 1.2 = 6.8 + 0.01N$$

New equilibrium:

$$32 - 0.02N = 6.8 + 0.01N$$

$$25.2 = 0.03N$$

$$N = 840$$

Wage:

$$W = 6.8 + 0.01(840) = 15.2$$

$$\boxed{N = 840, \quad W = 15.2}$$

(c) Change in wage:

$$\Delta W = 15.2 - 16 = -0.8$$

Change in employment:

$$\Delta N = 840 - 800 = 40$$

$$\boxed{\Delta W = -0.8, \quad \Delta N = 40}$$

(d) Interpretation:

Immigration increases labour supply, which lowers the equilibrium wage and raises employment. In this example, the wage falls by \$0.80 and employment rises by 40 workers.

Question 3. Immigrants and native-born workers as imperfect substitutes

Demand for native-born workers:

$$W = 40 - 0.01N_n + 0.005N_i$$

Supply of native-born workers:

$$W = 10 + 0.02N_n$$

(a) When $N_i = 200$:

Demand becomes:

$$W = 40 - 0.01N_n + 0.005(200)$$

$$W = 40 - 0.01N_n + 1 = 41 - 0.01N_n$$

Set equal to supply:

$$41 - 0.01N_n = 10 + 0.02N_n$$

$$31 = 0.03N_n$$

$$N_n = 1033.33$$

Wage:

$$W = 10 + 0.02(1033.33) = 30.67$$

$$\boxed{N_n = 1033.33, \quad W = 30.67}$$

(b) When $N_i = 300$:

Demand becomes:

$$W = 40 - 0.01N_n + 0.005(300)$$

$$W = 40 - 0.01N_n + 1.5 = 41.5 - 0.01N_n$$

Set equal to supply:

$$41.5 - 0.01N_n = 10 + 0.02N_n$$

$$31.5 = 0.03N_n$$

$$N_n = 1050$$

Wage:

$$W = 10 + 0.02(1050) = 31$$

$$\boxed{N_n = 1050, \quad W = 31}$$

(c) Interpretation:

Since immigrant labour enters positively in the native-born demand equation, immigrants are complements to native-born workers. As immigration rises, both native-born employment and wages increase.

Question 4. Migration decision and present value

(a) Net annual earnings advantage of Country B:

$$58,000 - 52,000 - 2,000 = 4,000$$

$$\boxed{\$4,000}$$

(b) Present value over 35 years:

$$PV = 4000 \left(\frac{1 - (1.04)^{-35}}{0.04} \right)$$

First compute:

$$(1.04)^{-35} \approx 0.253$$

So:

$$PV = 4000 \left(\frac{1 - 0.253}{0.04} \right)$$

$$PV = 4000 \left(\frac{0.747}{0.04} \right)$$

$$PV = 4000(18.675) = 74,700$$

$$\boxed{PV \approx \$74,700}$$

(c) Subtract moving cost:

$$74,700 - 15,000 = 59,700$$

Since this is positive, migration is worthwhile.

$$\boxed{\text{Migration is worthwhile}}$$

(d) If moving cost rises to \$30,000:

$$74,700 - 30,000 = 44,700$$

Still positive, so migration is still worthwhile.

Yes, migration is still worthwhile

Question 5. Immigrant earnings assimilation

(a) Initial earnings gap:

$$44,000 - 32,000 = 12,000$$

\$12,000

(b) Earnings gaps:

After 5 years:

$$47,000 - 39,000 = 8,000$$

After 10 years:

$$50,000 - 46,000 = 4,000$$

After 15 years:

$$54,000 - 52,000 = 2,000$$

8,000, 4,000, 2,000

(c) Gap closed after 15 years:

$$12,000 - 2,000 = 10,000$$

\$10,000

(d) Percentage of initial gap closed:

$$\frac{10,000}{12,000} \times 100 = 83.33\%$$

83.33%

Question 6. Point system calculation

(a) Total points:

$$21 + 20 + 11 + 8 + 0 + 7 = 67$$

67

(b) Since passing score is 67:

Yes, the applicant qualifies

(c) If language points improve by 5:

$$67 + 5 = 72$$

72

(d) Since the applicant already has 67, additional points needed:

0

Question 7. Labour force, unemployment rate, LFPR, and employment rate

- Population = 2,400,000
- Employed = 1,560,000
- Unemployed = 120,000

(a) Labour force:

$$1,560,000 + 120,000 = 1,680,000$$

1,680,000

(b) Unemployment rate:

$$\frac{120,000}{1,680,000} \times 100 = 7.14\%$$

7.14%

(c) Labour force participation rate:

$$\frac{1,680,000}{2,400,000} \times 100 = 70\%$$

70%

(d) Employment rate:

$$\frac{1,560,000}{2,400,000} \times 100 = 65\%$$

65%

Question 8. Labour force and unemployment rate

- Labour force = 1,800,000
- Unemployment rate = 7.5%

(a) Number unemployed:

$$0.075(1,800,000) = 135,000$$

$$\boxed{135,000}$$

(b) Number employed:

$$1,800,000 - 135,000 = 1,665,000$$

$$\boxed{1,665,000}$$

(c) Employment rises by 45,000 and unemployment falls by 45,000:

New unemployed:

$$135,000 - 45,000 = 90,000$$

New unemployment rate:

$$\frac{90,000}{1,800,000} \times 100 = 5\%$$

$$\boxed{5\%}$$

Question 9. Monthly labour market transitions

(a) Net monthly change in unemployment:

$$180,000 - 210,000 = -30,000$$

$$\boxed{-30,000}$$

(b) New stock of unemployed:

$$960,000 - 30,000 = 930,000$$

$$\boxed{930,000}$$

(c) Unemployment rate:

$$\frac{930,000}{16,000,000} \times 100 = 5.81\%$$

$$\boxed{5.81\%}$$

Question 10. Incidence and duration

$$UR = \text{Incidence} \times \text{Duration}$$

(a) Region A:

$$2.5\% \times 3 = 7.5\%$$

7.5%

(b) Region B:

$$1.5\% \times 5 = 7.5\%$$

7.5%

(c) Both regions have the same unemployment rate:

7.5%

(d) Interpretation:

The same unemployment rate can come from many short spells (high incidence, short duration) or fewer long spells (low incidence, long duration).

Question 11. Youth vs adults

(a) Youth:

$$6.0\% \times 2.0 = 12.0\%$$

Adults:

$$2.0\% \times 2.8 = 5.6\%$$

So the approximation holds exactly here.

Yes, it holds for both groups

(b) Higher incidence:

$$6.0\% > 2.0\%$$

Youth

(c) Longer duration:

$$2.8 > 2.0$$

Adults

(d) Interpretation:

Youth have high unemployment because they enter and leave jobs frequently.
Their unemployment spells are short, but incidence is high.

Question 12. Labour force, employment, and unemployment

- Population = 5,000,000
- LFPR = 66%
- Employment rate = 61%

(a) Labour force:

$$0.66(5,000,000) = 3,300,000$$

3,300,000

(b) Number employed:

$$0.61(5,000,000) = 3,050,000$$

3,050,000

(c) Number unemployed:

$$3,300,000 - 3,050,000 = 250,000$$

250,000

(d) Unemployment rate:

$$\frac{250,000}{3,300,000} \times 100 \approx 7.58\%$$

7.58%

Question 13. Discouraged workers and the unemployment rate

Initial:

- Labour force = 3,000,000
- Unemployed = 180,000

(a) Initial unemployment rate:

$$\frac{180,000}{3,000,000} \times 100 = 6\%$$

6%

(b) New unemployed:

$$180,000 + 30,000 = 210,000$$

210,000

(c) New labour force:

$$3,000,000 - 40,000 + 30,000 = 2,990,000$$

2,990,000

(d) New unemployment rate:

$$\frac{210,000}{2,990,000} \times 100 \approx 7.02\%$$

7.02%

(e) Interpretation:

The unemployment rate does not capture all hardship because discouraged workers who stop searching are removed from the labour force and no longer counted as unemployed.

Question 14. Long-term unemployment

(a) Country X:

$$6.0\% \times 10\% = 0.6\%$$

0.6%

(b) Country Y:

$$6.0\% \times 35\% = 2.1\%$$

2.1%

(c) More serious long-term unemployment problem:

Country Y

(d) Interpretation:

Even with the same overall unemployment rate, Country Y has a much larger share of workers unemployed for long periods, which suggests more serious structural labour market problems.

Question 15. Canada labour market indicators

- Unemployment rate = 5.8%
- LFPR = 65.5%
- Population = 31,000,000

(a) Labour force:

$$0.655(31,000,000) = 20,305,000$$

$$\boxed{20,305,000}$$

(b) Number unemployed:

$$0.058(20,305,000) = 1,177,690$$

$$\boxed{1,177,690}$$

(c) Number employed:

$$20,305,000 - 1,177,690 = 19,127,310$$

$$\boxed{19,127,310}$$

(d) Employment rate:

$$\frac{19,127,310}{31,000,000} \times 100 \approx 61.70\%$$

$$\boxed{61.70\%}$$